

Fact Sheet

X-37B ORBITAL TEST VEHICLE

June 2026

CONTACT: KATHLEEN BRETT
KBRETT@SWFOUND.ORG



SECURE
WORLD
FOUNDATION

SUMMARY

The X-37B OTV is a technology demonstrator and experimental vehicle which is likely used for testing reusable space launch vehicle technologies, such as guidance and thermal protection, and on-orbit testing of new sensor technologies and satellite hardware for risk reduction. The X-37B has also deployed small satellites. While it does have some potential capability for orbital inspection, repair, and retrieval, it is unlikely to perform these functions given its limited payload bay. It has near-zero feasibility as an orbital weapons system for attacking targets on the ground.

BACKGROUND

- X-37B is an experimental reusable spaceplane, similar to the US Space Shuttle but much smaller, completely robotic, and using more advanced technologies.¹
- X-37B is designed to be launched into space on top of a large rocket, stay in orbit for months to years, and then re-enter and land on a runway. It has thrusters for on-orbit maneuvering and de-orbit, but no engines for powered flight in the air: it is a glider in the atmosphere.
- X-37B started as a NASA program in 1999, but was transferred to DARPA in 2004. DARPA transferred it to the US Air Force (USAF) in 2006 after budget issues.²
- Hobbyist tracking indicates the X-37B typically operates at an inclination of 38-54.5 degrees and an altitude of 285 to 400 km (180 to 250 mi), although the seventh flight was tracked going into a highly elliptical orbit (HEO) out to 38,838 km.³
- The X-37B is an asset of the US Space Force (USSF)'s 3rd Space Experimentation Squadron, located at Schriever Space Force Base, Colorado.⁴ However, as of June 2024, the USSF's Space Delta 9, specifically the 5th Operations Squadron, is responsible for its launch, on-orbit operations, and landing.⁵



X-37B.

Image credit: US Air Force

X-37B ORBITAL FLIGHT HISTORY

Flight History Table

Launch Date	Launch Location	Landing Date	Landing Location	Time on Orbit	Flight
Apr. 22, 2010	Cape Canaveral Air Force Station, FL	Dec. 3, 2010	Vandenberg Air Force Base, CA	224 days	OTV-1
Mar. 5, 2011	Cape Canaveral Air Force Station, FL	Jun. 16, 2012	Vandenberg Air Force Base, CA	469 days	OTV-2
Dec. 11, 2012	Cape Canaveral Air Force Station, FL	Oct. 17, 2014	Vandenberg Air Force Base, CA	675 days	OTV-3
May 20, 2015	Cape Canaveral Air Force Station, FL	May 7, 2017	Kennedy Space Center, FL	718 days	OTV-4
Sep. 7, 2017	Kennedy Space Center, FL	Oct. 27, 2019	Kennedy Space Center, FL	780 days	OTV-5
May 17, 2020	Cape Canaveral Air Force Station, FL	Nov. 12, 2022	Kennedy Space Center, FL	908 days	OTV-6
Dec. 28, 2023	Kennedy Space Center, FL	Mar. 7, 2025	Vandenberg Space Force Base, CA	434 days	OTV-7
Aug. 21, 2025	Kennedy Space Center, FL	–	–	–	OTV-8

DEBATE OVER MISSION AND RATIONALE FOR THE X-37B PROGRAM

Official objectives of the X-37B program include “space experimentation, risk reduction and concept of operations development for reusable space vehicle technologies.”⁶ However, some have questioned the official mission and this has led to speculation that its “real” mission may be secretly an offensive weapons platform. The actual activities the X-37B has carried out indicate its mission is likely exactly what the USSF claims it is.

Over the course of its eight missions, the X-37B has served as a testbed for multiple technologies, including autonomous orbital landing, advanced Hall Effect thrusters,⁷ and oscillating heat pipes⁸ that have been used operationally in other programs. The X-37B has also served as a research platform and hosted experiments on material exposure and degradation, space radiation on seeds, and microwave power beaming. Multiple cubesats have also deployed on X-37B missions, although it is not always clear if they were deployed as rideshares from the upper stage or from the payload bay of the X-37B itself. Finally, the X-37B’s orbit has typically been in line with that of an imagery, surveillance, or reconnaissance (ISR) platform, such as repeating ground tracks and low altitudes. Given that the OTV-7 mission was at a much higher altitude of nearly 39,000 km, it is possible it was testing a new type of sensor or payload, but it is unclear what kind or why.

While there has been much speculation of the X-37B's role for interacting with other satellites on orbit or serving as a weapons platform, there is no evidence to date of this occurring. To date, the X-37B has never approached or rendezvoused with any other known space object and generally orbits far below the vast majority of operational satellites, with the exception of OTV-7, which was spotted by an amateur astronomer in February 2024 in a Highly Elliptical Orbit (HEO) of 323 x 38,838 km x 59.1 deg inclination.⁹

Physics further limits its utility as a space-to-ground weapon. The X-37B's payload bay and available power are very limited. Hyperkinetic weapons dropped from the bay would need to be equipped with thrusters capable of performing a significant deorbit burn, which is unlikely given the available room. The X-37B itself re-enters like the space shuttle landing at an estimated 200 mph (321 kph), which means it travels in the atmosphere much slower than a nuclear re-entry vehicle on a ballistic arc or a hyperkinetic weapon. It thus would need to carry conventional explosives to do any significant damage.¹⁰ The X-37B after re-entry would be a slow moving, not-very-maneuverable glide bomb, easy prey for any air defense system along its path to the target.

MAJOR ACTIVITIES OF X-37B

The first flight of the X-37B was in April 2010. OTV-1 was launched on a United Launch Alliance-built Atlas V Evolved Expendable Launch Vehicle. The program was described as a "flexible space test platform to conduct various experiments," and that it would "allow satellite sensors, subsystems, components and associated technology to be efficiently transported to and from the space environment."¹¹ It landed at Vandenberg Air Force Base, California, in December 2010 after 224 days in orbit.¹² The second flight of the X-37B, OTV-2 began in March 2011 with a launch via the Atlas V rocket from Cape Canaveral Air Force Station, Florida. Program officials again did not say anything specific about the mission but did anticipate that it would be able to stay on orbit for 270 days, given analysis of OTV-1's performance parameters.¹³ Around the 270-day mark of its flight, USAF officials announced that they were extending its mission to allow for "additional experimentation opportunities." It landed at Vandenberg AFB, California, in June 2012, after 469 days in orbit.¹⁴

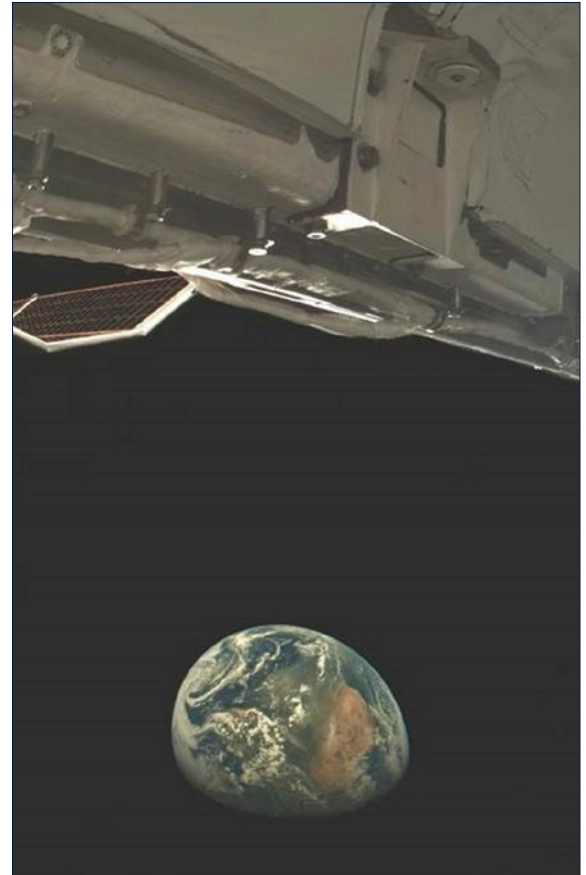
The third flight of the X-37B, OTV-3 was the first to reuse a flight vehicle, the OTV-1 in this case. The original launch was planned for October 2012, but due to a flight anomaly, the launch was pushed back.¹⁵ OTV-3 was launched in December 2012 via an Atlas 5 from Cape Canaveral Air Force Base, Florida, with a mission to "incorporate the lessons learned during the refurbishment process on OTV-1." It spent 675 days in orbit before landing in October 2014, at Vandenberg AFB, California.¹⁶ In May 2015, OTV-4 was launched via an Atlas 5 launch rocket from Cape Canaveral Air Force Station, Florida.¹⁷ Program officials stated that OTV-4 would carry a Hall thruster to experiment with this technology for its eventual use in the Advanced Extremely High Frequency (AEHF) military satellite program.¹⁸ NASA officials stated that OTV-4 would carry Materials Exposure and Technology Innovation in Space (METIS), a program intended to test almost 100 different types of materials to the space environment for over 200 days.¹⁹ OTV-4 landed at NASA's Space Shuttle Landing Site at the Kennedy Space Center, Florida, in May 2017, after 718 days in space.²⁰

In September 2017, OTV-5 launched from the Kennedy Space Center, Florida, via SpaceX's Falcon 9 reusable launch vehicle. It was announced that the mission would carry "small satellite rideshare" and an "Advanced Structurally Embedded Thermal Spreader payload to test experimental electronics and oscillating heat pipe technologies in the long duration space environment."²¹ On October 27, 2019, the OTV-5 flight of the X-37B completed an, at that time, record-breaking 780-day stay in orbit with a landing at NASA's Kennedy Space Center Shuttle Landing Facility.²²

OTV-6 featured a new service module added to the aft of the spacecraft which gave it more room for payloads and experiments.²³ One of these experiments was the satellite FalconSAT-8, built by students at the USAF academy.²⁴ OTV-6 also tested an on-orbit power beaming system (PRAM) which collected solar power and transformed it into a microwave beam. Russian sources claim that in October 2021, OTV-6 released a small object that spent a day about 200 meters away from OTV-6 before moving away entirely. OTV-6 landed in November 2022 after spending 908 days in orbit, which is the longest flight to date for the X-37B.²⁵

Prior to the December 2023 launch of the OTV-7, USSF officials said that tests of the X-37B would include “operating the reusable spaceplane in new orbital regimes, experimenting with future space domain awareness technologies, and investigating the radiation effects on materials provided by NASA.”²⁶ USSF officials further noted that they were able to evaluate how well the US Space Surveillance Network’s radars and telescopes tracked the vehicle, as well as monitor how other countries’ sensors locate and track OTV-7. In October 2024, the USSF announced that OTV-7 would undertake aerobraking maneuvers in order for it to reach a low enough orbit that it could safely discard its service module components, arguing that this would ensure compliance with recognized standards for space debris mitigation.²⁷ These aerobraking maneuvers were a new capability for the program. It is unclear how long the procedure took and whether the OTV-7 stayed in LEO after or if it returned to return to its planned experiments and testing in HEO. On March 7, 2025, the OTV-7 landed at Vandenberg Space Force Base, ending the seventh mission of the X-37B after 434 days in orbit.²⁸

On August 21, 2025, the most recent flight of the X-37B, OTV-8, was launched from the Kennedy Space Center on a Falcon 9 launch vehicle.²⁹ While an August 2025 USSF press release did not provide any updates regarding the OTV-8 mission, it expand on the contents of the entire program, stating that, “The X-37B is a dynamic and responsive spacecraft responsible for conducting a range of tests and experiments that expedite the development of critical next-generation technologies and operational concepts for reusable space capabilities.”³⁰ Boeing was tasked with building the OTV-8 and the company explained that OTV-8 “is hosting several technology demonstrations from government partners on this mission, including laser communications and a quantum inertial sensor designed to support navigation when GPS is unavailable;” these partners include the Air Force Research Laboratory (AFRL) and Defense Innovation Unit (DIU).³¹ As of May 2026, the OTV-8 is still in orbit.



X-37B OTV-7 in orbit.

Image credit: USSF

MANEUVERABILITY OF THE X-37B

There has also been significant speculation over the maneuverability of the X-37B while on orbit. The X-37B does have onboard thrusters that give it a large specific impulse, a measurement of a rocket engine's efficiency, and the X-37B has raised and lowered its orbit fairly often, particularly during OTV-3, OTV-4, and OTV-5, suggesting significant delta-v capability as well. Former Secretary of the Air Force Heather Wilson publicly suggested that the X-37B could also dip down into the atmosphere and use its wings to make unpredictable changes to its orbit.³² In February 2025, General Chance Saltzman, Chief of Space Operations, stated that the United States had announced it intended to use aerobraking maneuvers because, "We wanted to let everybody know that we were going to do it...We want other nations to observe it. We want to share the data."³³ Aerobraking maneuvers consist of a series of passes through the Earth's atmosphere to lower its orbit in a manner that uses less fuel, for the purpose of getting low enough orbit to safely discard service module components. To complete the aforementioned aerobraking maneuvers, the vehicle had to undergo modifications, including development of a new collision avoidance system.³⁴

TRANSPARENCY CONCERNS

Much of the speculation and consternation over the X-37B stems from the project's secretive nature. The mission and activities of the X-37B have long been classified, and the program's budget is currently unknown. The US military openly applauds the program for being a significant success, but does not provide much information about its on-orbit activities. This may indicate it is part of a covert intelligence program, but it may also indicate the testing of offensive technologies or capabilities. From the mission's launch to February 2025, the USSF had identified 1.7 million collision avoidance opportunities, though it is unclear as to what that means.³⁵

SECRET DEPLOYMENT OF SATELLITES

The X-37B secretly deployed three small satellites during its fifth mission between 2017 and 2019. Although the USAF stated before launch that OTV-5 would deploy small satellites, it was unclear if they were rideshares deployed from the upper stage or deployed from the X-37B payload bay itself.³⁶ The actual deployment time of the three satellites was not announced and is still uncertain. The three satellites were not included in the public satellite catalog maintained by the US military until after OTV-5 landed. The United States did not provide any orbital positional information on OTV-5 or the three small satellites it deployed. The United States reported OTV-5 to the United Nations in conformity with the Convention on Registration of Objects Launched into Outer Space on September 17, 2018, one year after launch and on the same timeline as past X-37B flights.³⁷ This withholding of information raises questions about US commitment to international norms and transparency and confidence building measures, as well as the purpose of these satellites.

ENDNOTES

1. "X-37B Orbital Test Vehicle Fact Sheet," Air Force Space Command, April 17, 2015, <http://www.af.mil/About-Us/Fact-Sheets/Display/Article/104539/x-37b-orbital-test-vehicle/>.
2. "X-37B Orbital Test Vehicle Fact Sheet," Air Force Space Command, *ibid*.
3. Jonathan McDowell, "Jonathan's Space Report No. 703," October 17, 2017, <http://planet4589.org/space/jsr/back/news.703>.
4. "Next X-37B Orbital Test Vehicle Scheduled to Launch," US Space Force, May 6, 2020, <http://www.spaceforce.mil/News/Article/2177702/next-x-37b-orbital-test-vehicle-scheduled-to-launch>.
5. "US Space Force Activates New Squadron to Oversee X-37B Orbital Test Vehicle," *SOFREP News Team*, June 8, 2024, <https://sofrep.com/news/us-space-force-5th-sops/>.
6. "Boeing-built X-37B Orbital Test Vehicle Successfully Completes 3rd Flight," Boeing, October 17, 2014, <https://boeing.mediaroom.com/2014-10-17-Boeing-built-X-37B-Orbital-Test-Vehicle-Successfully-Completes-3rd-Flight>.
7. Jeff Foust, "X-37B lands after record-setting mission," *SpaceNews*, May 7, 2017, <https://spacenews.com/x-37b-lands-after-record-setting-mission/>.
8. Craig Covault, "USAF to Launch First Spaceplane Demonstrator," *Aviation Week Science and Technology*, August 3, 2008, <http://bit.ly/cvnkRA>.
9. Jonathan McDowell, "Congrats to Tomi Simola for locating the secret X-37B spaceplane. OTV 7 is in a 323 x 38838 km x 59.1 deg orbit. Could be testing out a new HEO IR sensor for future early warning satellites - just a wild speculation on my part here." *Twitter*, February 9, 2024, <https://twitter.com/planet4589/status/1755974823119872007?s=20>.
10. Jeff Foust, "X-37B lands after record-setting mission," *SpaceNews*, May 7, 2017, <https://spacenews.com/x-37b-lands-after-record-setting-mission/>.
11. Vanessa Young, "New unmanned spacecraft set to launch," *Air Force News Service*, April 16, 2010, <https://www.af.mil/News/Article-Display/Article/116970/new-unmanned-spacecraft-set-to-launch/>.
12. "X-37B Orbital Test Vehicle lands at Vandenberg AFB," *Air Force News Service*, December 10, 2010, <https://www.af.mil/News/Article-Display/Article/114795/x-37b-orbital-test-vehicle-lands-at-vandenberg-afb/>.
13. Ammani Lyle, "Air Force launches second unmanned spacecraft," Secretary of the Air Force Public Affairs, March 5, 2011, <https://www.patrick.spaceforce.mil/News/Article-Display/Article/330116/air-force-launches-second-un-manned-spacecraft/>.
14. Stephen Clark, "Military space shuttle receives mission extension," *Spaceflight Now*, November 29, 2011, <https://www.spaceflightnow.com/news/n1111/29otv2/>; "X37-B Orbital Test Vehicle lands at Vandenberg," *30th Space Wing Public Affairs*, June 16, 2012, <https://www.vandenberg.spaceforce.mil/News/Article-Display/Article338649/x-37b-orbital-test-vehicle-lands-at-vandenberg/>.
15. Amy McCullough, "X-37B Launch Delayed," *Air Force Magazine*, October 25, 2012, <https://web.archive.org/web/20140220190719/http://www.airforcemag.com/DRArchive/Pagers/2012/October%202012/Octover%2024%202012/X-37BLaunchDelayed.aspx>.
16. Eric Badger, "Air Force launched 3rd X-37B Orbital Test Vehicle," *Air Force News Service*, December 11, 2012, <https://www.af.mil/News/Article-Display/Article/109994/air-force-launch-es-3rd-x-37b-orbital-test-vehicle/>; Stephen Clark, "U.S. military's X-37B space plane lands in California," *Spaceflight Now*, October 17, 2014, <https://spaceflightnow.com/atlas/av034/141017landing/>.
17. Justin Ray, "X-37B embarks on fourth voyage in orbit," *Spaceflight Now*, May 20, 2015, <https://spaceflightnow.com/2015/05/20/recap-story-x-37b-embarks-on-fourth-voyage-in-space/>.
18. David Szondy, "Hall ion thrusters to fly on X-37B spaceplane," *New Atlas*, April 29, 2015, <https://newatlas.com/us-air-force-x-37b-hall-thrust-er/37200/>.
19. Justin Ray, "NASA gives more information on its experiment aboard the X-37B," *Spaceflight Now*, May 6, 2015, <https://spaceflightnow.com/2015/05/06/nasa-gives-more-information-on-its-experiment-abroad-the-x-37b/>.
20. Jeff Foust, "X-37B lands after record-setting mission," *SpaceNews*, May 7, 2017, <https://spacenews.com/x-37b-lands-after-record-setting-mission/>.

21. "Air Force preparing to launch fifth Orbital Test Vehicle mission," *Air Force News Service*, August 31, 2017, <https://www.af.mil/News/Article-Display/Article/1296566/air-force-preparing-to-launch-fifth-orbital-test-vehicle-mission/>; "45th SW launches 5th OTV mission," *Air Force News Service*, September 7, 2017, <https://www.af.mil/News/Article/1301794/45th-sw-launches-5th-otv-mission/>.
22. Secretary of the Air Force Public Affairs, "X-37B Breaks Record, lands After 780 Days In Orbit," United States Air Force, October 27, 2019, <https://www.af.mil/News/Article-Display/Article/1999734/x-37b-breaks-record-lands-after-780-days-in-orbit/>.
23. Joseph Revithick, "X-37B's Power Beaming Payload A Reminder Of Potential Orbital Microwave Anti-Satellite Weapons," *TheDrive.com*, May 19, 2020, <https://www.thedrive.com/the-war-zone/33531/x-37bs-power-beaming-payload-arereminder-of-potential-orbital-microwaveanti-satellite-weapons>.
24. Dmitry Stefanovich, "Sometimes VPK still delivers," *Twitter*, December 8, 2021, <https://twitter.com/KomissarWhipla/status/1468593293235793924?s=20>; Vladimir Kozin, "Cold Star War: The US has Questions for Moscow about Space, Russia has even more Questions for the US," *VKP-News*, November 29, 2021, <https://vpk-news.ru/articles/64859>.
25. Stefano D'Urso and David Cenciotti, "Reflecting On The X-37B's Latest Record-Breaking Mission," *The Aviationist*, December 24, 2022, <https://theaviationist.com/2022/12/24/reflecting-on-the-x-37bs-latest-record-breaking-mission/>.
26. "Department of the Air Force scheduled to launch seventh X-37B mission," *Air Force News Service*, November 8, 2023, <https://www.spaceforce.mil/News/Article-Display/Article/3583347/department-of-the-air-forcescheduled-to-launch-seventh-x-37b-mission/>; Marco Langbroek, "X-37B begins novel space maneuver," *SatTrackCam Leiden (b)log*, February 13, 2024, <https://sattrackcam.blogspot.com/2024/02/imaging-x-37b-robotic-spaceplane.html>.
27. Sandra Erwin, "X-37B to jettison payload module in orbital shift, Space Force announces," *SpaceNews*, October 10, 2024, <https://space-news.com/u-s-military-spaceplane-to-perform-orbital-maneuvers/>.
28. Secretary of the Air Force Public Affairs, "X-37B Orbital Test Vehicle concludes seventh successful mission," *Air Force News Service*, March 13, 2025, <https://www.spaceforce.mil/News/Article-Display/Article/4112259/x-37b-orbital-test-vehicle-concludes-seventh-successful-mission/>; <https://www.spaceforce.mil/News/Article-Display/Article/4112259/x-37b-orbital-test-vehicle-concludes-seventh-successful-mission/>.
29. Secretary of the Air Force Public Affairs, "US Space Force launches eighth X-37B mission," *Air Force News Service*, August 22, 2025, <https://www.spaceforce.mil/News/Article-Display/Article/4283172/us-space-force-launches-eighth-x-37b-mission/>.
30. Secretary of the Air Force Public Affairs, "US Space Force launches eighth X-37B mission," *ibid*.
31. "Boeing-Built X-37B Spaceplane Launches, Beginning Eighth Mission," *Boeing Press Release*, August 22, 2025, <https://boeing.mediaroom.com/2025-08-22-Boeing-Built-X-37B-Space-plane-Launches,-Beginning-Eighth-Mission>.
32. Secretary of the Air Force Public Affairs, "X-37B Orbital Test Vehicle concludes seventh successful mission," United States Space Force, March 13, 2025, <https://www.spaceforce.mil/News/Article-Display/Article/4112259/x-37b-orbital-test-vehicle-concludes-seventh-successful-mission/>.
33. Vivienne Machi, "How The X-37B Is Shaping The Future Of The U.S. Space Force," *Aviation Week Network*, February 21, 2025, <https://aviation-week.com/space/budget-policy-regulation/how-x-37b-shaping-future-us-space-force>.
34. Machi, "How The X-37B Is Shaping The Future Of The U.S. Space Force," *ibid*.
35. Machi, "How The X-37B Is Shaping The Future Of The U.S. Space Force," *ibid*.
36. "Air Force preparing to launch fifth Orbital Test Vehicle mission," *ibid*.
37. United States of America, "Information Furnished in Conformity with the Convention on Registration of Objects Launched into Outer Space," United Nations Office For Outer Space Affairs, September 17, 2018, <https://www.unoosa.org/oosa/osoindex/data/documents/us/st/stsgser.e859.html>.



525 Zang Street, STE. D
Broomfield, CO 80021 USA
v: + 1 303 554 1560

1779 Massachusetts Ave. NW
Washington, DC 20036 USA
v: + 1 202 568 6212

e: info@swfound.org